DE HAVILLAND AIRCRAFT OF CANADA LIMITED

TEMPORARY REVISION

PSM: 1-64-2 **AIRCRAFT: DHC-6** **SERIES: 400**

PUBLICATION: MAINTENANCE MANUAL **TEMP. REVISION NO.: TR57-1**

Insert in 57-00-00 before Page 1

WINGS - GENERAL DATA

REASON FOR ISSUE:

To introduce S.O.O. 6478, Wing Hardpoint Installation, NATO, STA 228.75 and 266.25.

ACTION:

In Chapter 57, Section 57-00-00, ignore existing Para 1 and insert new para as shown on Page 2 of this Temporary Revision.

-Continued overleaf-

RECORDING INSTRUCTIONS:

Record this TEMPORARY REVISION NO. and the relevant information in the appropriate columns of the TEMPORARY REVISION INDEX provided at the front of the manual.

Wings – General Data

1. General

The wing group consists of the wings, nacelles, wing struts and wing fences. A detachable wing tip is installed on each outer wing. The wings are provided with attachment fittings for the flap/aileron system at the trailing edge, and bleed air piping.

The wing structure consists of front, main, and rear spars, ribs, and skins reinforced with spanwise corrugations bonded to the skins. The nacelle structure is riveted to the top and bottom surfaces of the wing. A front and rear attachment fitting is provided on the root end of the wing, and a wing strut attachment fitting is located on the main spar. Hinge fittings for the wing flaps and aileron systems are carried at intervals along the rear spar, and a flap shroud is riveted to the spar. Nose ribs mounted forward from the front spar support the fixed leading edge skin which extends outboard from the nacelle structure. Inboard of the nacelle structure, the leading edge is hinged to provide access to engine control runs. Access panels in the bottom skin give access to the wing flap and aileron mechanism.

Aircraft incorporating Mod S.O.O. 6247 have integral fuel tanks installed in the leading edge of each wing. Each tank occupies three bays between the four outer leading edge ribs, forward of the main spar.

On aircraft with lightning protection (Mod 6/1272), a horn assembly is attached to each wing tip, and protection straps and a conduit assembly are installed between the wing tip and wing structure.

For aircraft incorporating Mod 6/2351, the wing tips have rubber washers installed. Torque the screws to compress the rubber washers (which are underneath the metal washers) to approximately one-half of their original thickness. For Pre Mod 6/2351 wing tips, without rubber washers, torque the #10 screws to 10 – 16 in–lbs.

The engine compressor wash (Mod S.O.O. 6050 or Mod 6/1316), a tube assembly with spray holes is installed in each nacelle structure.

Aircraft incorporating Mod S.O.O. 6478 have provisions to mount external stores beneath the wings at STA 228.75 and STA 266.25. When no external store is installed on hardpoint attachment provision locations, ensure that AN7 bolts are installed as follows:

- Apply MIL-PRF-23827 Type II (Aeroshell Grease 7) or equivalent to barrel nuts per PSM 1-64-2 Ch. 12-20-10
- Install AN7 bolts and torque to 270-300 in-lb
- Dome seal MIL-PRF-81733 Type II (P/S 870 Class B) bolt head and hardware per PSM 1-64-2 Ch. 20-50-01

Note Mounting any external stores onto these provisions requires a separate airworthiness approval and must comply with the requirements of Interface Control Drawing (ICD) C6IC1003.

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2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Wing (Mod 6/1752)

1) Remove wing root, wing strut and nacelle fairings. Raise wing inboard leading edge. Remove upholstery in cabin roof.

2) Trestle tail and opposite wing.

3) Remove engine. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

4) Disconnect five wing-to-fuselage electrical connectors. Disconnect generator cable assembly at power distribution box and remove clamps in cabin roof. Disconnect and tag remaining electrical wiring at wing-to-fuselage joint. Withdraw cable and wiring clear of wing joint.

5) Disconnect aileron cables at turnbuckles in cabin roof.

6) Disconnect fuel shut-off, throttle, and propeller pitch engine control cables at turnbuckles adjacent to wing nose spar.

7) If wing de-icer boots are fitted, loosen clamps and remove hose from two de-icing lines at wing root. Cap open connections.

8) Unbutton insulation blanket at bleed air line and check connector for freedom of movement. Loosen clamps and remove hose from fuel vent line. Cap all open connections.

9) Remove flap control rod seal at wing root. Remove cotter pin, nut, washer and bolt at flap control rod. Remove cotter pin, nut and washer attaching flap hinge arm to fitting on fuselage.

10) Attach sling SD12505 to wing and take weight on crane. (Refer to 07-00-00, Lifting and Shoring – General Data).

11) Disconnect flexible fuel line from elbow at top of wing strut. Remove oil breather pipe adjacent to wing strut, disconnecting pipe at wing strut on aircraft embodying Mod 6/1326. Cap open lines.

12) Remove cotter pin and slacken nut at wing strut to fuselage attachment. If Mod S.O.O. 6247 is embodied, withdraw wing tank system electrical wiring from wing strut. Support wing strut, remove cotter pin, nut, washer, and bolt at wing strut to wing attachment and lower wing strut to rest on ground.

    Note If required, wing strut may be removed completely by disconnecting flexible fuel pipe at elbow and removing nut, washer and bolt at wing strut/fuselage attachment.

13) Remove cotter pins, washers, pins and the rear spar wing-to-fuselage attachment pin at the rear wing/fuselage attachment and nut, washer, bolt and shims at front wing/fuselage attachment. Note position and thickness of shims used between front attachments fittings for reinstallation. Carefully move wing outboard and withdraw cables at joint. Ensure bleed air pipe is withdrawn from connector and flap hinge arm is released from fitting on fuselage.

14) Lower wing onto transportation cradle.

B. Install Wing (Mod 6/1752)

1) Attach sling SD12505 to wing and hoist wing into position (refer to 07-00-00, Lifting and Shoring – General Data). Ensure flap hinge arm engages fuselage fitting, and bleed air pipe slides into connector.

2) Apply light coating of corrosion preventive compound (refer to 20-30-01, Protective Coatings Hard and Soft Film Corrosion Preventive Compound – General Data), to plain portion of wing/fuselage attachment bolts. Ensure threads of bolt are dry.

3) Line up wing/fuselage attachment fittings and insert bolt using the front acorn SD12514. Insert shims at front attachment only, between fuselage beam and rear lug of wing fitting. Shims to be sufficient to take up end float with wing pushed back as far as possible, and a minimum gap of 0.010 inch at rear attachment between fuselage beam and front and rear lugs of wing fitting.

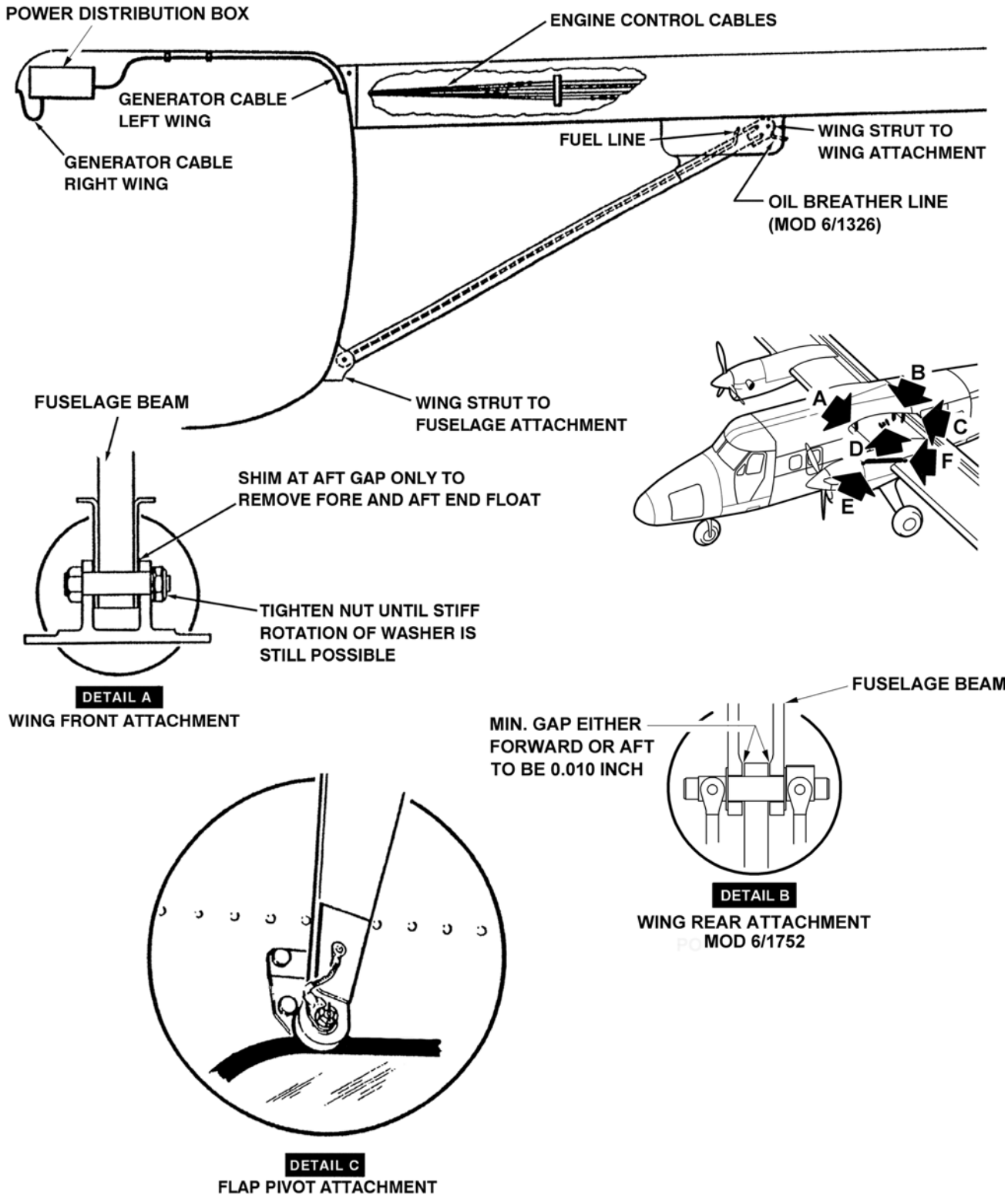
4) At front attachment bolt, ensure that washer installed below nut and painted face of adapter lug are both clean; tighten nut until stiff rotation of washer by fingers is still possible. Install the rear spar wing-to-fuselage attachment pins with washers on outside of fuselage fitting. Install pin, washers and cotter pins in both ends of rear spar wing-to-fuselage attachment pin.

5) Swing wing strut into position at wing attachment and install bolt, washer, nut and cotter pin. Tighten nut at wing strut/fuselage attachment and install cotter pin. If Mod S.O.O. 6247 is embodied, feed wing tank system electrical wiring through wing strut, and reconnect at bottom of strut.

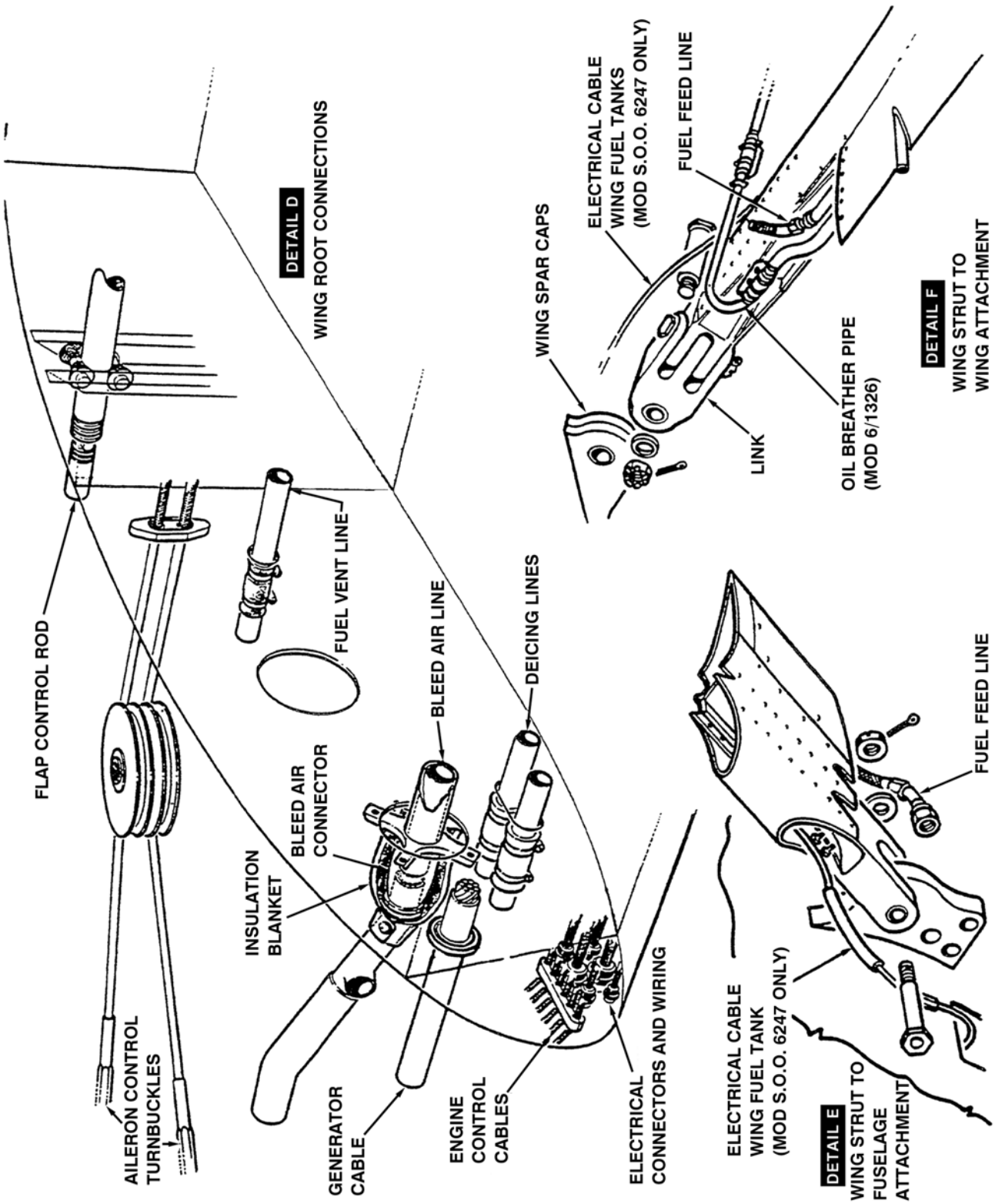
    Note If wing strut was removed completely, install bolt, washer, nut and cotter pin at wing strut/fuselage attachment. Connect flexible fuel line to elbow at bottom of strut.

6) Remove sling SD12505 from wing.

7) Connect flexible fuel line to elbow at top of wing strut. Install oil breather pipe adjacent to wing strut. Tighten clamps and secure bonding.



Wing Installation
Figure 1 (Sheet 1 of 2)



Wing Installation
Figure 1 (Sheet 2 of 2)

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8) Engage flap push-pull rod connection and install bolt, washer, nut and cotter pin. Install washer, nut and cotter pin attaching flap hinge arm to fitting on fuselage. Install flap control rod seal at wing root.

9) Install hose connecting fuel vent line at wing root and tighten clamps. Check bleed air connector at wing joint, and button insulation blanket over connector. Examine fume seals and replace if unserviceable.

10) If wing de-icer boots are fitted, install hose and tighten clamps on two de-icing lines at wing root.

11) Connect turnbuckles of fuel shut-off, power and propeller control cables at wing nose spar.

12) Connect turnbuckles of aileron cables in cabin roof.

CAUTION **IF BOTH LEFT AND RIGHT AILERON CONTROLS HAVE BEEN DISCONNECTED, ENSURE THAT AILERON CABLES AT AILERON CONTROL QUADRANT IN CABIN ROOF ARE CORRECTLY INSTALLED. (REFER TO Figure 2, DETAIL B).**

13) Connect electrical wiring, wing-to-fuselage. Route and connect generator cable assembly to power distribution box and install clamps in cabin roof. Connect five wing-to-fuselage electrical connectors.

14) Install engine (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data). Rig fuel shut-off, power and propeller controls (refer to 76-10-00, Power Control – Maintenance Practices).

15) Remove tail and wing trestles.

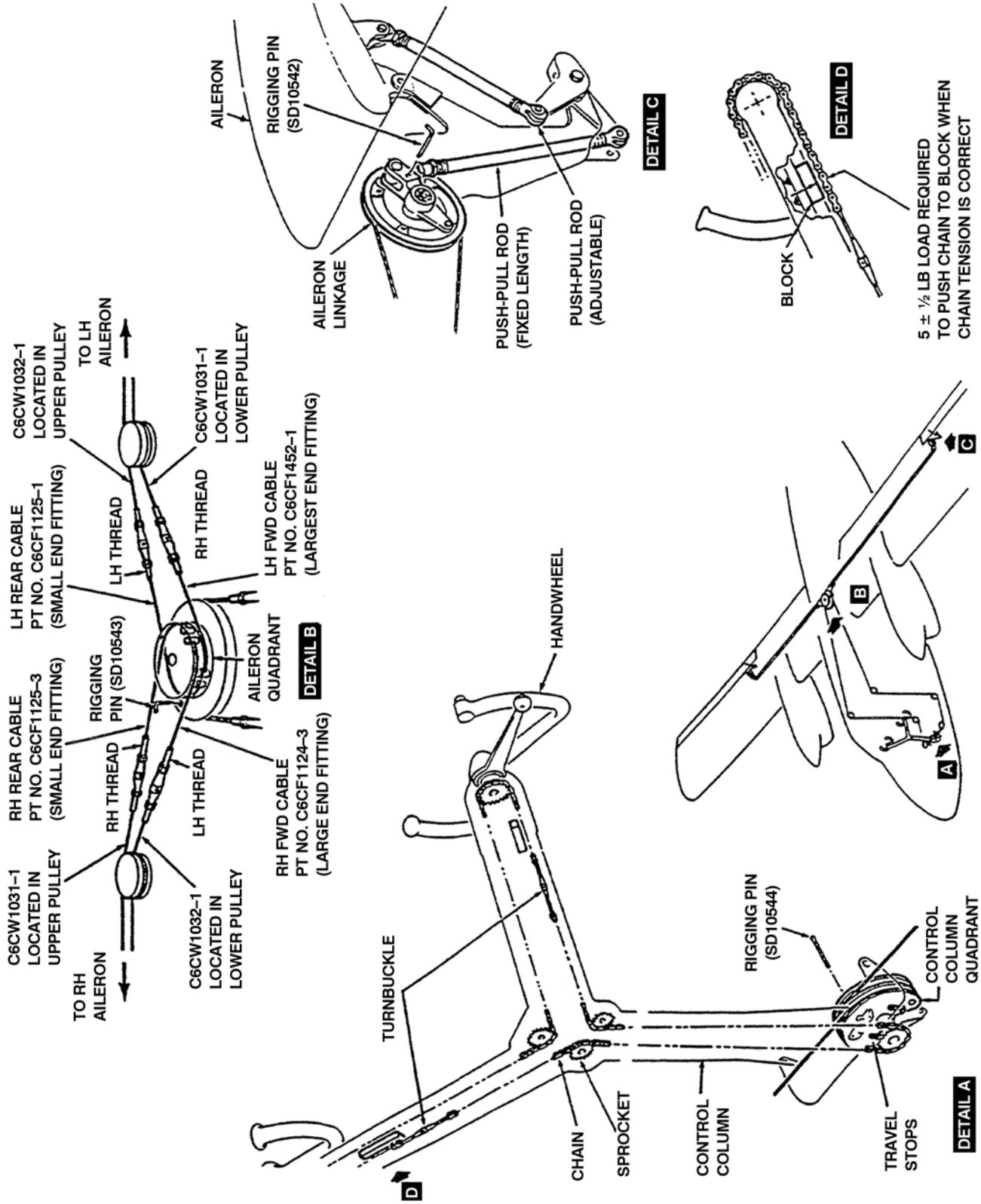
16) Check rigging of wing flaps, tension aileron control cables and rig ailerons and aileron tabs. (Refer to 27-10-00, Aileron Control System – General Data).

17) Function electrical systems to engine and wing affected by removal and installation. With booster pumps operating, check fuel lines for leaks.

18) Install upholstery in cabin roof. Lower and secure wing inboard leading edge. Install nacelle, wing strut and wing root fairings.

Note For aircraft incorporating Mod 6/2340, the fairings have rubber washers installed, torque the screws to compress the rubber washers (which are underneath the metal washers) to approximately one-half of their original thickness. For fairings without rubber washers (Pre Mod 6/2340) torque the #10 screws to 10 – 16 in–lbs.

19) During engine run, check bleed air system. (Refer to 21-40-11, Bleed Air Shut-off Valve – Maintenance Practices).



Aileron System – Rigging
Figure 2

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